Electrical Transient Analyzer Program

Short-Circuit Analysis

IEC 60909 Standard

3-Phase, LG, LL, & LLG Fault Currents

	Swing	V-Control	Load	Total
Number of Buses:	1	0	1	2

	XFMR2	XFMR3	Reactor	Line/Cable/ Busway	Impedance	Tie PD	Total
Number of Branches:	1	0	0	0	0	0	1

	Synchronous Generator	Power Grid	Synchronous Motor	Induction Machines	Lumped Load	Total
Number of Machines:	0	1	0	0	0	1

System Frequency:	50.00
Unit System:	Metric
Project Filename:	Short_Circuit_Analysis
Output Filename:	C:\Users\Eng Abdalsamie\Desktop\ETAP - Projects\Short Circuit Analysis\Fault Analysis Project\ETAP 19 Project\Sort Circuit.SI2S

Adjustments

Tolerance	Apply Adjustments	Individual /Global	Percent
Transformer Impedance:	Yes	Individual	
Reactor Impedance:	Yes	Individual	
Overload Heater Resistance:	No		
Transmission Line Length:	No		
Cable / Busway Length:	No		

Temperature Correction	Apply Adjustments	Individual /Global	Degree C
Transmission Line Resistance:	Yes	Individual	
Cable / Busway Resistance:	Yes	Individual	

Bus Input Data

Bus					Initial Voltage	
ID	Type	Nom. kV	Base kV	Sub-sys	%Mag.	Ang.
Bus1	SWNG	33.000	33.000	1	100.00	0.00
Bus2	Load	11.000	11.000	1	100.00	-30.00

2 Buses Total

All voltages reported by ETAP are in % of bus Nominal kV.
Base kV values of buses are calculated and used internally by ETAP.

2-Winding Transformer Input Data

Transformer	Rating					Z Variation			% Tap Setting		Adjusted	Phase Shift	
ID	MVA	Prim. kV	Sec. kV	% Z	X/R	+ 5%	- 5%	% Tol.	Prim.	Sec.	% Z	Type	Angle
T1	45.000	33.000	11.000	5.00	5.00	0	0	0	0	0	5.00	Dyn	30.00

2-Winding Transformer Grounding Input Data

Transformer	Rating			Grounding								
				Conn.	Primary				Secondary			
ID	MVA	Prim. kV	Sec. kV	Type	Type	kV	Amp	ohm	Type	kV	Amp	ohm
T1	45.000	33.000	11.000	D/Y					Solid			

Branch Connections

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVAb			
ID	Type	From Bus	To Bus	R	X	Z	Y
T1	2W XFMR	Bus1	Bus2	2.21	11.06	11.28	

Power Grid Input Data

		Rating		% Impedance 100 MVA Base			Grounding
Power Grid ID	Connected Bus ID	MVASC	kV	R	X"	R/X"	Type
U1	Bus1	55.000	33.000	14.61531	181.22980	0.08	Wye - Solid

Total Connected Power Grids (= 1): 55.000 MVA

SHORT- CIRCUIT REPORT

Fault at bus: **Bus2**
Nominal kV = 11.000
Voltage c Factor = 1.10 (User-Defined)

Contribution		3-Phase Fault		Line-To-Ground Fault					Positive & Zero Sequence Impedances Looking into "From Bus"			
From Bus	To Bus	% V	kA	% Voltage at From Bus			kA Symm. rms		% Impedance on 100 MVA base			
ID	ID	From Bus	Symm. rms	Va	Vb	Vc	Ia	3I0	R1	X1	R0	X0
Bus2	Total	0.00	2.734	0.00	94.90	95.82	3.994	3.994	1.83E+001	2.10E+002	2.21E+000	1.11E+001
Bus1	Bus2	5.87	2.734	54.70	110.00	56.29	3.994	3.994 *	1.83E+001	2.10E+002	2.21E+000	1.11E+001
U1	Bus1	110.00	0.911	110.00	110.00	110.00	0.769	0.000	1.61E+001	1.99E+002	1.61E+001	1.99E+002

		3-Phase	L-G	L-L	L-L-G
Initial Symmetrical Current (kA, rms)	:	2.734	3.994	2.367	4.419
Peak Current (kA), Method C	:	6.862	10.027	5.943	11.094
Breaking Current (kA, rms, symm)	:		3.994	2.367	4.419
Steady State Current (kA, rms)	:	2.734	3.994	2.367	4.419

\# Indicates a fault current contribution from a three-winding transformer.
* Indicates a zero sequence fault current contribution (3I0) from a grounded Delta- Y transformer.

Short-Circuit Summary Report

3-Phase, LG, LL, LLG Fault Currents

Bus		3-Phase Fault			Line-to-Ground Fault				Line-to-Line Fault				*Line-to-Line-to-Ground			
ID	kV	I"k	ip	Ik	I"k	ip	Ib	Ik	I"k	ip	Ib	Ik	I"k	ip	Ib	Ik
Bus2	11.000	2.734	6.862	2.734	3.994	10.027	3.994	3.994	2.367	5.943	2.367	2.367	4.419	11.094	4.419	4.419

All fault currents are in rms kA. Current ip is calculated using Method C.

* LLG fault current is the larger of the two faulted line currents.

Sequence Impedance Summary Report

Bus		Positive Seq. Imp. (ohm)			Negative Seq. Imp. (ohm)			Zero Seq. Imp. (ohm)			Fault Zf (ohm)		
ID	kV	Resistance	Reactance	Impedance	Resistance	Reactance	Impedance	Resistance	Reactance	Impedance	Resistance	Reactance	Impedance
Bus2	11.000	0.22130	2.54600	2.55560	0.22130	2.54600	2.55560	0.02677	0.13383	0.13648	0.00000	0.00000	0.00000